

Operating Procedures

0.1 Pre-Flight Checks

1. Ardupilot parameter list + Mission waypoints
2. Shake the wing to avoid pinching servo cables with the bayonet
3. Check the fuselage tube connector while inserting in wing-box
4. Insert carefully wing connectors and check for pin retraction
5. Alcool on ESC
6. Clean Lidar surface
7. **Check** servo connector correctly inserted
8. Tail connector fastened
9. Condition check
 - (a) Visual inspection of the plane (intact, clean and working components):
 - i. Landing gear (LG)
 - ii. Structural integrity
 - iii. Fuselage (no loose components)
 - iv. Payload release correctly retained
 - v. Clean aerodynamic surfaces
 - vi. No holes in Oracover
 - (b) Rear capote sensors and antennas integrity check
 - (c) Front capote electronics integrity check
 - (d) Securely tightened bolts, retention mechanisms and propeller
 - (e) Check connectors condition and electronic components correctly fastened
 - (f) Wing retention pin inserted
 - (g) Correct movement of all moving parts:
 - i. Servomotors
 - ii. Control surfaces
 - iii. Payload release system
 - iv. Nose landing gear
10. Batteries check (SOC and correct retention):
 - (a) Radio battery
 - (b) Ground Control Station (GCS) / PC battery
 - (c) Main battery (main switch off)
11. CG check through balances for correct stability margin

0.2 Startup Procedures

1. Main switch connected
2. **SD card inserted**
3. GCS connected
4. GCS correctly functioning
5. Sensor calibration:
 - (a) IMUs completed
 - (b) Compass completed
 - (c) GPS completed
 - (d) Pitot completed
6. Check for NO errors from Ardupilot
7. Correct GPS signal
8. Correct LIDAR reading
9. FC core temperature check
10. ESC temperature check
11. SOC FC battery
12. SOC main battery from Radio
13. Correct autopilot output direction on control surfaces for safe flight (check if outputs are inverted)
14. Mission planner mission upload completed
15. Check written waypoints **one-by-one !**
16. Check written parameters
17. Correct first waypoint selected
18. Failsafe test completed
19. Temperature check ESC + FC
20. Pitot cover and other red-tagged items removed
21. Engine properly working

Take off Procedure: *Flaps* shall be set to the *first-notch (+1) position*. The take-off manoeuvre shall be performed by applying *full throttle* (80 A of current) while simultaneously applying *right rudder* to counteract the torque generated by the engine. The control stick shall only be pulled back after reaching V_{S1} (IAS) and preferably before exceeding the take-off box distance limitation of 10 m.

Cruise Procedure: The pilot shall stabilize the aircraft at either the best-range airspeed or the minimum sink rate speed. Trim of the elevator might be necessary depending on the loading condition of the UAV.

Payload Release Procedure: The payload shall be released as close as possible to the designated GPS coordinates. The pilot shall remain vigilant during this phase for potential release failures. Slight pitch variations might occur performing the manoeuvre, but should be well controllable, as the payload's centre of gravity is designed to be nearly coincident with the aircraft's structural centre of gravity.

Landing Procedure: The pilot shall reduce the airspeed below V_{FE} , extend *full flaps* (+2), and align the aircraft with the runway. The throttle shall be used solely to stabilize the descent. The pilot should aim to touch down at the beginning of the designated landing box.